



FACULTY OF INFORMATION TECHNOLOGY AND ELECTRICAL ENGINEERING

IoT Project Report

Moisture Level Detection System

By

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1. Motivation:

World's available fresh water resources are long considered abundant, and various users exploited them somewhat indiscriminately; however, in present, competition has arisen between different economic sectors for access to this resource. This particularly holds true in regions, where the need for timely, sufficient, and affordable good quality water for crops and animals has led agricultural producers to become increasingly sensitive and responsive to water issues.

Good irrigation water management will increase yields, improve crop quality, conserve water, save energy, decrease fertilizer requirements, and reduce non-point source pollution. Using soil moisture measurements is one of the best and simplest ways to get feedback to help make improved water management decisions. Moreover, the monitoring system of soil level detection will help farmers/maintainers to monitor plant requirements and build insights by analyzing gathered data. However, the installation, calibration, and interpretation of the data from these instruments is often overwhelming for most busy growers. Here's an attempt to provide practical recommendations for using these sensors to improve their operations.

2. Design and Architecture:

The project can work in a way where the farmer will know if whether the plants need watering or not. For large scale projects, where a huge area of land is to be irrigated, a system maintainer and a system assembler supervisor are also introduced in the model. Details can be seen in the use case diagram below;

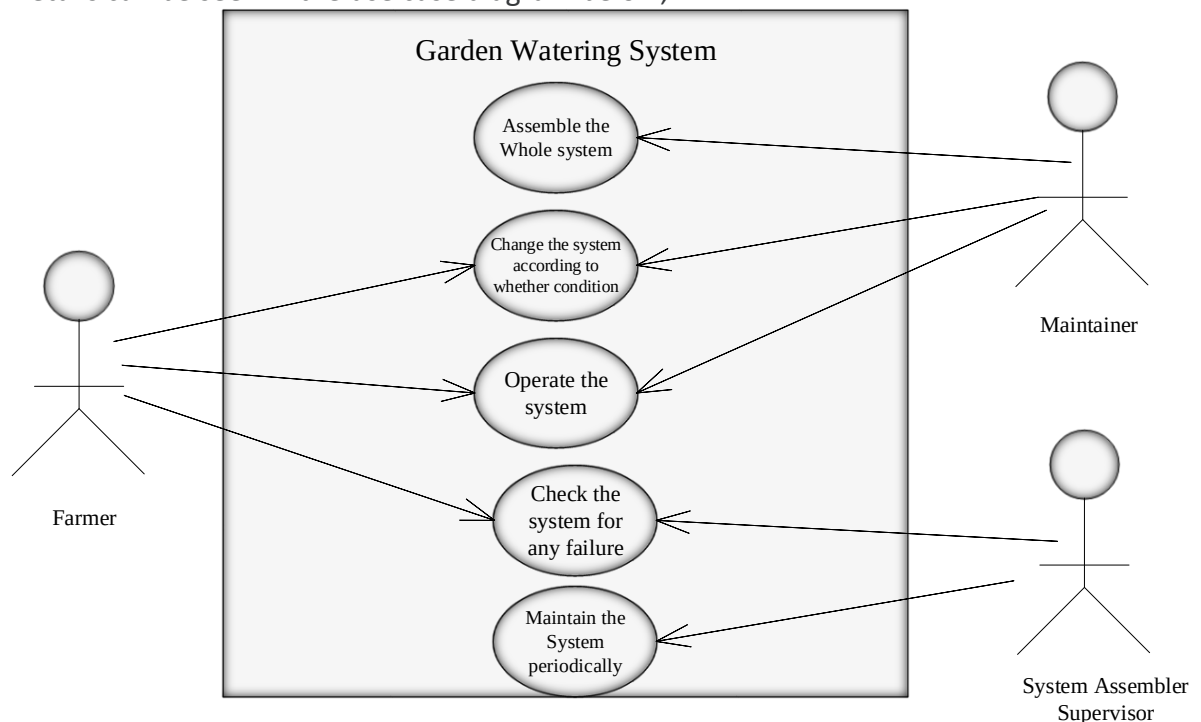


Figure: 2.1 Use Case Diagram

To obtain the results, we started from the very basic implementation which is explained below.

First system will be initialized, the sensor will collect the data from the soil and indicate the humidity on LED. If yellow light is on, it indicates that time to water the plant and if orange light is on, it indicates plant does not require watering. Later, the data is stored in the database and readings can be visualized in the form of graphs.

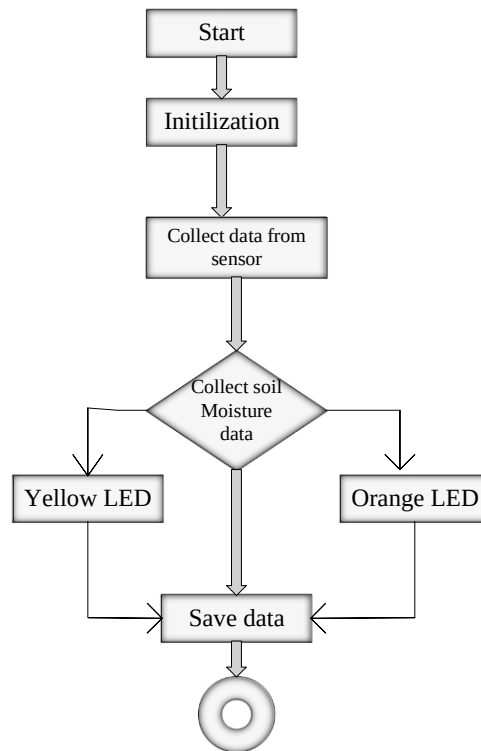


Figure: 2.3 Process Flow (part- a)

After achieving the above results, we extended it further. The project is extended in such a way that the monitoring system itself monitors and inform users about the needs of a plant. The extended implementation works as follows.

The collected data from sensor is stored in database and is visualized using graphical form representation on a designed dashboard. The dashboard will keep updating itself with defined intervals. If the moisture level of soil detects a specified threshold, an alert will be sent to user.

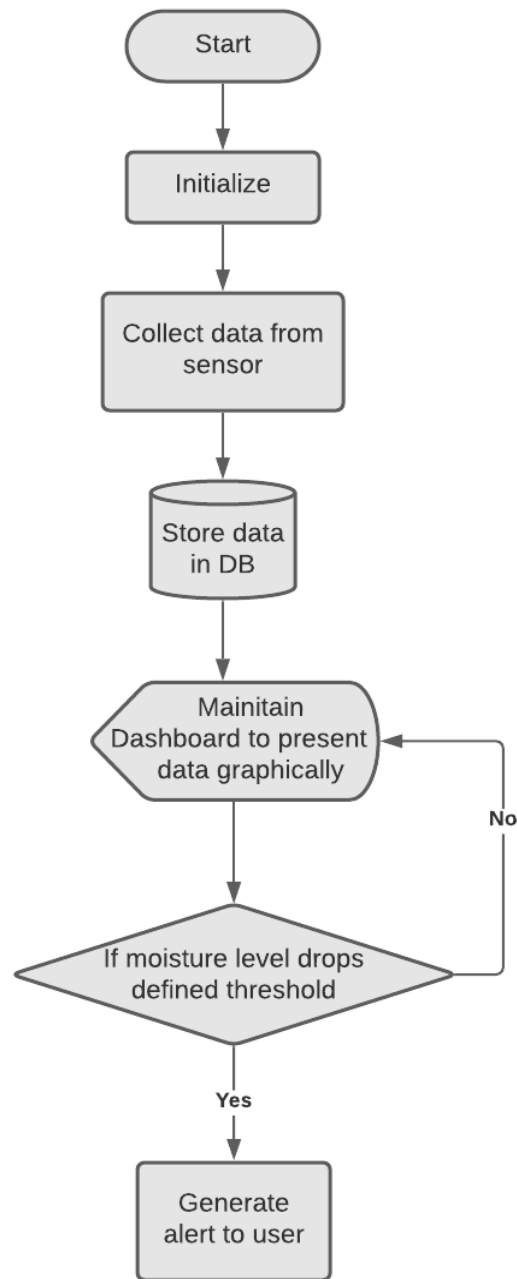


Figure: 2.3 Process Flow (part- b)

3. Implementation:

System is composed of the following

- Soil Moisture Sensor B105
- Arduino Uno
- InfluxDB for storage & visualization

Python script gathers values from the sensors using pymata4 library and ingest the values in Influx DB. As influx DB is a timeseries database, it is used to store runtime values.

The implementation is done using Python language. The python script is responsible for collecting the sensors data via Arduino using python library pymata4 and storing the data in influxDB.

1. Initiate `db_settings()` by
 - a. Initializing `influxDB` client
 - b. instantiate the `WriteAPI` and `QueryAPI`
2. initiate board settings by
 - a. creating `pymata4` instance
 - b. set the `set_pin_mode_analog_input` and `set_pin_mode_digital_output` for the board
3. while true:
 - a. `soil_level, event_change_timestamp = read values from board; analog_read()`
 - b. if `previous event change timestamp != event change timestamp`:
 - i. generate `Point()` attributes
 - ii. write the `Point` synchronously in database; using `write_api()` method

Figure 3.1 Pseudocode of python script

InfluxDB dashboard fetches data from the database and visualize them as line charts. For this purpose, InfluxDB bucket is queried upon by filtering the tags to retrieve the desired results.

The InfluxDB dashboard image depicts the timeseries graph where the plant's need can be visualized. User can select refreshing rate of the graph as per own choice. Moreover, user can customize the date range to be visualized in the dashboard.



Figure 3.2 Dashboard

The email alert is generated to notify user about the soil level moisture drop when soil level drops to specific threshold. For this purpose, 3rd party services i.e., mailgun is used to send alert emails as a “Plant Care Service”

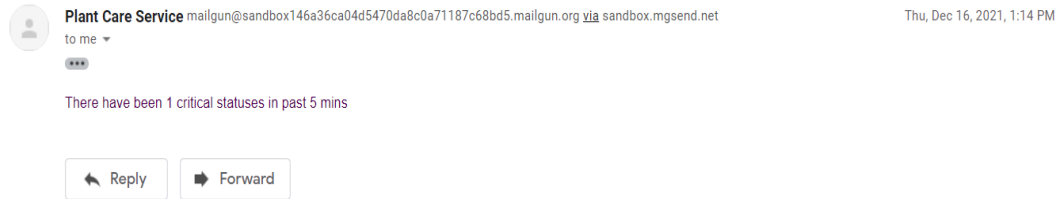


Figure: 3.3 Email Alert/Notification

4. Evaluation/Limitations:

The implementation results were achieved successfully and were satisfactory. Though there are limitations exists which are discussed below.

For now, the application is visualizing data only from one sensor, but it can be expended to a vast scale to automate agricultural processes.

As the application can be extended on a larger, the main limitation of the design includes that the more the area, the greater the number of sensors is required. More sensors mean more cost of installation. Moreover, automatic watering sensors could have been added so that the plants could be watered automatically but that will drastically increase the installation costs as price of automatic watering sensors and taps is a bit high.

Additionally, there can be added more advanced analytical features for instance using Machine learning algorithms to build insights for the longer-term changes.

Furthermore, security concerns have not been taken care of while collecting data from the sensors.

5. Discussion on Future Impacts:

Similar solutions include automatic gardening systems for smart homes, but they are yet available in fully automated homes, and companies charge a hefty amount for it. Moreover, some greenhouse farming houses also use similar sensors to keep track of the moisture system, but no such solution has been applied to irrigate a vast area of land by now. Our system possesses the functionality of integrating as many sensors as required, can be implemented for a huge area of land that needs irrigation, and thus has a business potential solution that can be developed and implemented.

As the system is used to monitor the plants' requirements, a trend book can be built by

analyzing the collecting data. By building the trends, intelligent decisions can be withdrawn which can help in plants' growth, improve efficiency of the system and saving critical time and energy. It can open new doors to study and implement features according to the plants need in nurseries/farms. For instance, trends can help in determining what hours of the day, the plant needs more water e.g., during sunlight etc. and thus this process can be automated to watering plant. It can also help in determining which factors/parameters are affecting plants' life as the requirements varies from plant to plant. So, automation can be done depending on the plant type.